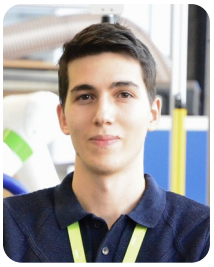
# Suavi Yildirim

suavi.yildirim@fau.de | linkedin.com/in/suaviyildirim



## Summary

**Mechanical Engineer and Master's candidate** with 5+ years of experience in robotics and automation, in the final semester of the M.Sc. Autonomy Technologies at FAU.

## Education

**Friedrich-Alexander-Universität Erlangen-Nürnberg** — Erlangen, DE
**Master of Science, Autonomy Technologies** — 04/2024 – Expected 03/2027

- Coursework completed; Master's thesis at Institute FAPS (submission end of October 2026)
- Thesis: Pneumatic levitation of wires using CFD simulation and machine-learning regression

**Ozyegin University** — Istanbul, TR
**Bachelor of Science, Mechanical Engineering** — 09/2017 – 07/2022

- 100% Merit Scholarship, Honor Student
- Graduated 1st in Mechanical Engineering (GPA 3.14/4.00, 1.8 German grade)

## Experience

**Institute for Factory Automation and Production Systems (FAPS)** — 07/2024 – Present
**Graduate Research Assistant (HiWi)** — Nürnberg, DE

- Coordinated development and validation of an automated pneumatic wire transport system
- Integrated Siemens S7-1500 PLCs, HMI panels, and sensors in TIA Portal; tuned control parameters

**Cobro Robot Technologies** — 09/2023 – 05/2024
**Concept Design Engineer – PreSales Robotics Automation** — Istanbul, TR

- Translated customer requirements into robotic automation concepts for industrial B2B applications
- Developed a flexible feeder system for vision-guided handling of randomly oriented parts

**Arche Robotics** — 07/2020 – 08/2023
**Mechanical Design Engineer – Autonomous Robotics** — Istanbul, TR

- Technical team lead for autonomous mobile robots, from design to commissioning at customer sites
- Delivered two fully operational AMR systems in MediaMarkt stores

## Projects & Awards

**Siemens Healthineers Robotics & AI Hackathon** | *Teach the Robot (VLA)* — 10/2026

- Robotics & simulation engineer (team of 6): simulation-based robot-learning loop for a bimanual Franka FR3 handover in Isaac Sim and Genesis; **12× faster** demo generation

**Siemens x Sony Immersive Design Challenge** | *BatteryTwin-XR* — 12/2024 – 04/2025

- Led Team NextCycle (XR digital twin for EV batteries); **1st place** of 900 participants

**OzU Rover Team** | *Ozyegin University* — 09/2017 – 04/2020

- Team lead (2018–2020); 13th of 56 teams at ERC 2019; two conference publications

## Volunteering & Extracurricular Activities

- **Siemens Campus Connect Student Ambassador** (since 05/2025); **Board Member**, FAU Space Gene
- **Panelist and speaker** on engineering and competitions; event support at Institute FAPS

## Technical Skills

**CAD & Simulation:** Siemens NX, CATIA V5, Mechatronics Concept Designer, SolidWorks, CFD, Isaac Sim
**Automation:** Siemens S7-1500 / TIA Portal, FANUC robots, ROS
**Programming:** Python, C++, MATLAB, OpenCV
**Manufacturing:** 3D printing (FDM, SLA), CNC basics, electronics prototyping
**Languages:** English (professional working proficiency), German (B2.1)

Neumarkt i.d.OPf., October 27, 2026 — Suavi Yildirim

# SUAVI YILDIRIM

**M.Sc. AUTONOMY TECHNOLOGIES, FAU ERLANGEN-NÜRNBERG**
**B.Sc. MECHANICAL ENGINEERING, OZYEGIN UNIVERSITY**

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# PORTFOLIO

Selected engineering projects in robotics, automation and product design — from mechanical design to simulation and robot learning.

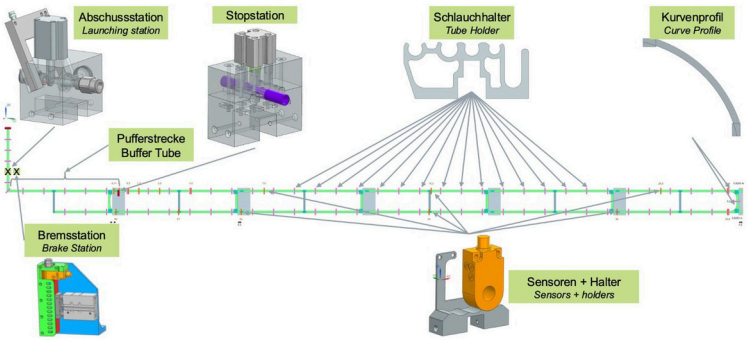
Online portfolio with videos and more pictures: **suavi-yildirim.github.io/portfolio/**


| PROJECT | ORGANISATION / PERIOD | HIGHLIGHT | PAGE |
|---|---|---|---|
| PNEUMATIC WIRE TRANSPORT (WTE) | FAU – Institute FAPS<br>since 07/2024 | **Stable wire transport at 20 m/s** | 3 |
| WIRE FLOATING EXPERIMENT (WFE) | FAU – Institute FAPS<br>2025 – 2026 | **Closed-loop wire levitation test rig** | 3 |
| NEXT2OEM – AUTOMATED WIRE HARNESS PRODUCTION | FAU – Institute FAPS × Audi<br>2024 – 2026 | **Tooling for the Audi harness demonstrator** | 4 |
| WIRE TWISTING UNIT | FAU – Institute FAPS | **Motor-driven wire twisting module** | 4 |
| ROBOTICS & AI HACKATHON – TEACH THE ROBOT | Siemens Healthineers<br>10/2026 | **12× faster demo generation** | 5 |
| IMMERSIVE DESIGN CHALLENGE | Siemens × Sony<br>12/2024 – 04/2025 | **1st place of 900 participants** | 5 |
| TORSIONAL MEASUREMENT DEVICE | 2E Design<br>2023 – 2024 | **Anti-roll bar stiffness tuning** | 6 |
| SPRING STIFFNESS MEASUREMENT DEVICE | 2E Design<br>2023 – 2024 | **Spring rates for suspension tuning** | 6 |
| SUSPENSION CONTROL ARM OF A RALLY CAR | 2E Design<br>2023 – 2024 | **Stronger, more durable control arm** | 6 |
| GRIPPER DESIGN FOR FOOD PACKAGING | Cobro Robot Technologies<br>2023 – 2024 | **+20% throughput, −90% cost** | 7 |
| COBOT PALLETIZER | Cobro Robot Technologies<br>2023 – 2024 | **Won follow-up projects** | 7 |
| FLEXIBLE FEEDER SYSTEM | Cobro Robot Technologies<br>2023 – 2024 | **Cost-effective mixed-part feeding** | 7 |
| AT1 MOBILE PLATFORM | Arche Robotics<br>2020 – 2023 | **AMR platform for 50 kg payload** | 8 |
| PROMOBOT | Arche Robotics<br>2020 – 2023 | **Deployed at MediaMarkt, 400 customers/month** | 8 |
| PURA | Arche Robotics<br>2020 – 2023 | **Cleans the air of 300 people per hour** | 9 |
| GENERATIVE DESIGN STUDY | Arche Robotics<br>2020 – 2023 | **30% lighter part** | 9 |
| ADDITIVE MANUFACTURING PART MANAGEMENT | Arche Robotics<br>2020 – 2023 | **Standardized print quality** | 9 |
| SURGICAL ROBOT FOR AUTONOMOUS SUTURING | OzU Robotics Lab | **TÜBİTAK 1001 research project** | 10 |
| KUKA KR20-3 LIGHT PAINTING | Ozyegin University – RobArt<br>2021 – 2022 | **Robot light painting** | 10 |
| KUKA KR20-3 PEN PLOTTER | Ozyegin University – RobArt<br>2021 – 2022 | **Robot pen-plotter art** | 10 |
| THE HOPE | OzU Rover Team<br>2019 – 2020 | **Top 36 of 102 teams (URC 2020)** | 11 |
| THEIA | OzU Rover Team<br>2018 – 2019 | **13th of 56 teams (ERC 2019)** | 11 |

## PNEUMATIC WIRE TRANSPORT (WTE)

**FAU – Institute FAPS** · since 07/2024 · Graduate research assistant

Siemens NX | 3D printing | Pneumatics






### What?

- As part of **Project SIMON** at the **Institute FAPS**, I contributed to an **automated pneumatic "shooting" system** that replaces manual **wire handling** and removes assembly bottlenecks for flexible wire components.

### How?

- Used **Siemens NX** for CAD modeling of the **experimental setup**, including tube holder, stop and brake stations.
- Used **FDM and SLA 3D printing** for parts with different **structural** and high-tolerance **aerodynamic** requirements.

### Results

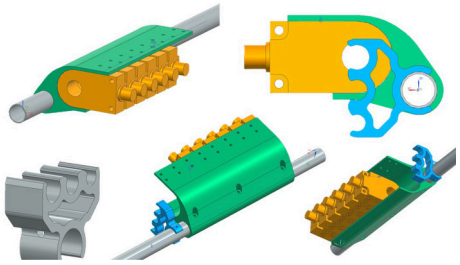
- **Achieved stable transport** at **20 m/s** while keeping a defined state of order for flexible leads.
- **Quantified** empirical pressure-loss coefficients for industrial-scale pneumatic system modeling.

## WIRE FLOATING EXPERIMENT (WFE)

**FAU – Institute FAPS** · 2025 – 2026 · Graduate research assistant, Master's thesis

S7-1500 | TIA Portal | PID | CFD







### What?

- Vertical test rig in which an **upward airflow holds a wire** in floating equilibrium – the basic case of every lifting section in **pneumatic wire transport**.

### How?

- Designed launching station, tube and **sensor holders** (Siemens NX, 3D printing).
- **Control cabinet**: S7-1500 software PLC, ET 200SP, IO-Link, Festo proportional pressure regulator and flow sensor.
- **PID control** of the inlet pressure from 5 inductive sensors; HMI for automated test runs.

### Results

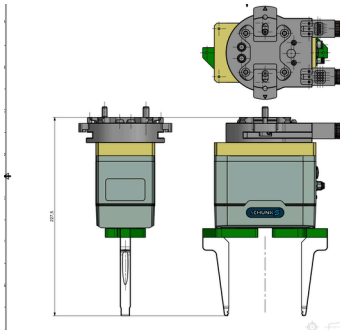
- **Reproducible** floating tests for wires of **0.5–3.5 m** and **0.5–6 mm²**.
- Dataset for the Master's thesis: **CFD** validation and **machine-learning** prediction of floating conditions.

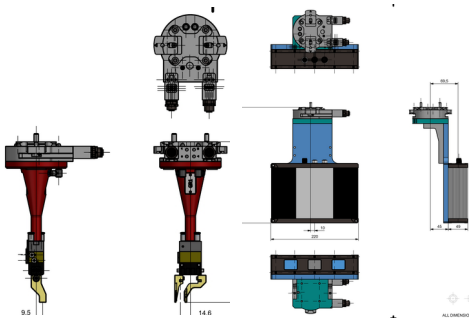
## NEXT2OEM – AUTOMATED WIRE HARNESS PRODUCTION

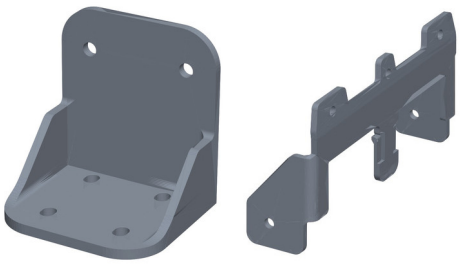
FAPS Lehrstuhl für Fertigungsautomatisierung und Produktionssystematik Prof. Dr.-Ing. Jörg Franke

**FAU – Institute FAPS × Audi** · 2024 – 2026 · Graduate research assistant

Wire harness | KBL | XR | Siemens NX | 3D printing







### What?

- **NEXT2OEM** is an **Audi-led** research project (BMWK, 2023–2026) with FAPS and industry partners.
- Goal: **digitise and automate** the whole wire-harness value chain, from connector to vehicle assembly, to bring harness production back to Germany.

### How?

- Helped **build the demonstrator**: fixtures, electric and pneumatic **gripper fingers** and a **camera mount** for the robot stations (Siemens NX).
- **Redesigned Audi parts** for 3D printing (FDM, PETG/PLA).
- Prepared **KBL harness data** (VDA 4964) for an **XR-based assembly guidance** system.

### Results

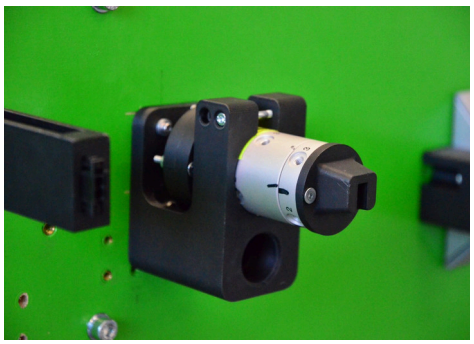
- Gripper and camera tooling **integrated** in the demonstrator stations.
- **Print-ready** Audi parts for fast iterations of the setup.
- Project result: Audi validated the prototype on a centre-console harness in Ingolstadt.

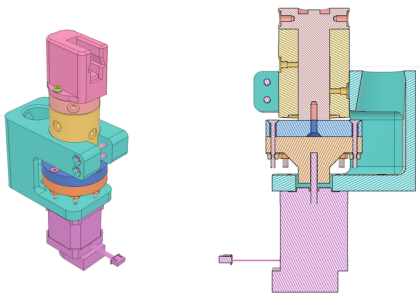
## WIRE TWISTING UNIT

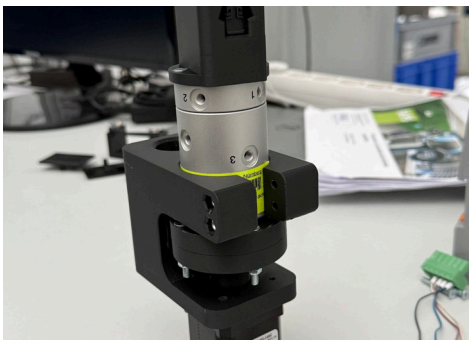
FAPS Lehrstuhl für Fertigungsautomatisierung und Produktionssystematik Prof. Dr.-Ing. Jörg Franke

**FAU – Institute FAPS** · Design and build

Wire harness | Stepper drive | Arduino | 3D printing







### What?

- A compact **twisting unit** for the **Robotik Challenge** setup: it holds the wire end in a **connector holder** and turns it to **twist the wire**.

### How?

- Designed the **bracket, motor coupling and connector holder** around an igus stepper motor and a rotary feedthrough.
- **Closed-loop stepper** drive (CL42T) with an **Arduino Nano**; speed tests up to 600 rpm.
- Printed parts for fast iterations; mounts directly on the task board.

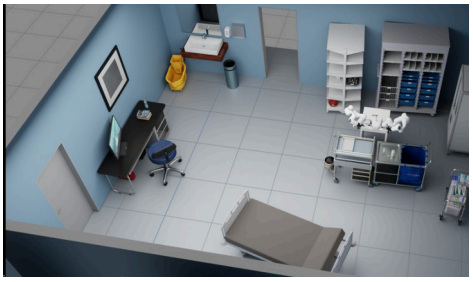
### Results

- **Built and tested** unit, integrated in the Robotik Challenge task board.
- Makes **wire twisting** – a typical wire-harness process step – available as a robot task.

## ROBOTICS & AI HACKATHON – TEACH THE ROBOT

**Siemens Healthineers** · 10/2026 · Robotics & Simulation Engineer, team of 6

Isaac Sim | Genesis | LeRobot | VLA | Quest 3







### What?

- 3-day **Siemens Healthineers Robotics & AI Hackathon 2026** in Erlangen: build the **learning loop** for a **vision-language-action (VLA)** robot companion in the cathlab.
- Task: bimanual box handover with a **Franka FR3 Duo** in simulation.

### How?

- **Robotics & Simulation Engineer** in a 6-person team.
- Built a **scripted IK expert** in **NVIDIA Isaac Sim 6.0** that generates demos with moved box poses (16/18 successful).
- Fine-tuned $\pi_{0.5}$ **(LoRA)** and **ACT**; rebuilt the task in **Genesis** for parallel data generation.

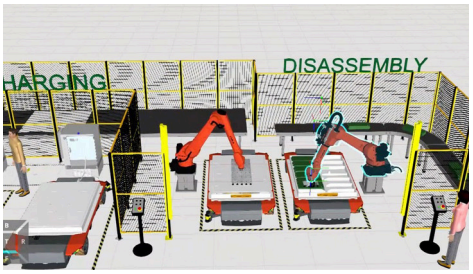
### Results

- Showed that the **data, not the model**, was the bottleneck: the 94 provided demos contain one single motion.
- **12× faster** demo generation: from 6 min to about 30 s per demo.
- Prototyped **VR teleoperation with Meta Quest 3** for human demos.

## IMMERSIVE DESIGN CHALLENGE

**Siemens × Sony** · 12/2024 – 04/2025 · Team lead

XR | Digital Twin | NX











### What?

- The Immersive Design Challenge is a **global engineering competition** organized by Siemens Digital Industries Software in collaboration with Sony.
- Encourages **sustainable design** by bringing immersive technologies such as **XR and digital twins** into real engineering problems.
- Scale: ~**900 students**, **230+ universities**, **38 countries**.

### How?

- **Led** the team **NextCycle**.
- **Developed** BatteryTwin-XR: a Siemens-integrated **digital twin** for a sustainable **EV battery lifecycle**.
- **Designed** an XR framework with the Sony XR head-mounted display to support **battery monitoring, reuse and recycling** decisions.
- Focused on **circular economy**: better recycling efficiency, less battery waste.

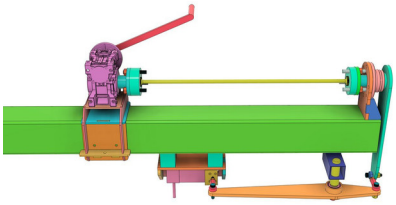
### Results

- **Won 1st place** among the international finalist teams.
- Recognized by Siemens and Sony for an **innovative, sustainability-driven** XR solution.
- Awarded a **Sony XR headset** and a 1-year **NX Immersive Designer** license for academic and research use.
- Invited to **Siemens Realize LIVE**; helped expand immersive design capabilities at FAU.

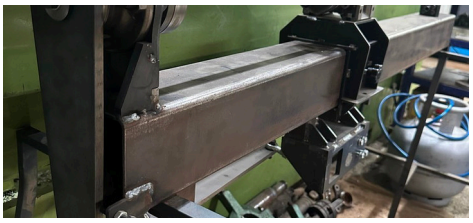
## TORSIONAL MEASUREMENT DEVICE

**2E Design** · 2023 – 2024 · Mechanical design

Test rig | Load cell | Motorsport







### What?

- **Developed** a **torsional measurement** device for analysing the mechanical characteristics of **anti-roll bars** in automobiles.

### How?

- **Implemented** a hub and bearing setup to assess **anti-roll bar twisting**.
- **Transferred** the load through lever arms to a **1-ton capacity** load cell for accurate measurement.

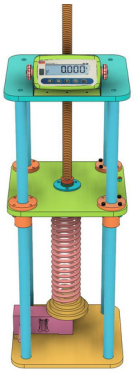
### Results

- The device allowed **measurement and adjustment** of anti-roll bar twist **stiffness**, ensuring optimal performance for the targeted **racing** vehicles.

## SPRING STIFFNESS MEASUREMENT DEVICE

**2E Design** · 2023 – 2024 · Mechanical design

Test rig | Motorsport



### What?

- **Developed** a **spring stiffness measurement** device for analysing springs used in racing **car suspension systems**.

### Results

- The device enabled accurate measurement of **spring coefficients**, allowing the suspension to be tuned for **optimal track performance**.

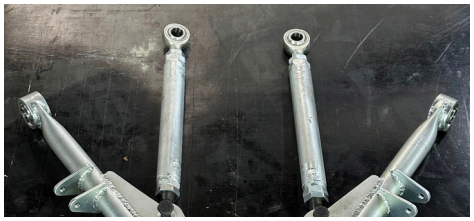


## SUSPENSION CONTROL ARM OF A RALLY CAR

**2E Design** · 2023 – 2024 · Mechanical design

Redesign | Manufacturing | Motorsport








### What?

- Conducted extensive **R&D** to improve the **suspension control arm** of a **rallycross** vehicle, to optimise performance and durability under demanding racing conditions.

### How?

- **Redesigned** the suspension arm after the **original broke** during a race.
- **Manufactured** a more **rigid and robust** arm and integrated it into the vehicle.

### Results

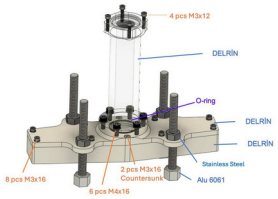
- The new design gave a **stronger, more durable** suspension arm, clearly improving **vehicle performance** and reliability on the track.

## GRIPPER DESIGN FOR FOOD PACKAGING

**Cobro Robot Technologies** · 2023 – 2024 · Concept design engineer

ABB Delta | Vacuum gripping








### What?

- **Developed** robot grippers for **ABB Delta robots**, tailored to the operational and hygiene requirements of a high-speed **food production company**.

### How?

- Designed grippers for high-speed **pick-and-place** operations.
- **Optimised gripper mass** to meet the robot payload limits.
- **Used O-ring sealing** for reliable, repeatable suction performance.

### Results

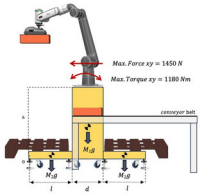
- **Integrated** into the production line: **20% higher** packaging throughput.
- **90% cost reduction** by replacing standard components with custom suction cups.

## COBOT PALLETIZER

**Cobro Robot Technologies** · 2023 – 2024 · Concept design engineer

ABB CRB 15000 | Cobot | Structural analysis








### What?

- **Developed** a **mobile palletizing robot** with collaborative robots that moves easily between **production lines** for more flexibility and efficiency.

### How?

- Performed **static and dynamic analyses** to design a robust base fixture.
- Selected the **ABB CRB 15000** cobot with a **12 kg payload**.

### Results

- The system was **successful** and helped the company win further projects, as it stands out from stationary palletizers on the market.

## FLEXIBLE FEEDER SYSTEM

**Cobro Robot Technologies** · 2023 – 2024 · Concept design engineer

Machine vision | Bin picking







### What?

- A **flexible feeding system** for manufacturers that need robotic automation to **separate mixed products**.

### How?

- A camera above the feeder locates **randomly oriented parts**; the robot picks them (**vision-guided handling**).

### Results

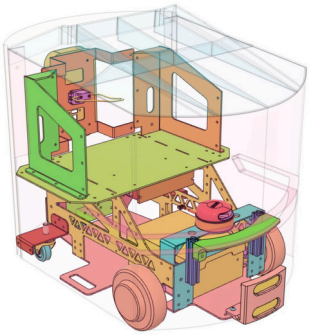
- A **cost-effective** automated feeder that sorts mixed parts, ready to be extended with **image processing and AI**.

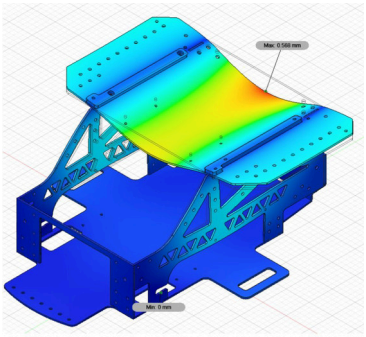
## AT1 MOBILE PLATFORM

ARCHE ROBOTICS

**Arche Robotics** · 2020 – 2023 · Mechanical design engineer

SolidWorks | Fusion 360 | FEA







### What?

- **AT1** is an indoor **autonomous mobile robot** platform (AMR).
- A versatile platform for **multiple applications**.
- Designed the chassis for a **50 kg** payload.

### How?

- Used **SolidWorks** and **Fusion 360** for the mechanical design.
- Ran an **FEA study** of the forces and moments on the chassis.
- 3 mm **6061 aluminium** sheet-metal parts joined by TIG welding.

### Results

- **Launched** AT1, now used from **healthcare** to **retail** and **hospitality**.
- Basis for ongoing development of new hardware and software modules.

## PROMOBOT

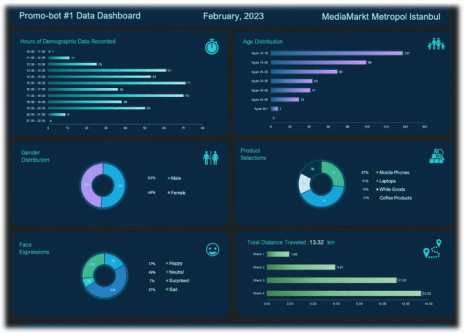
ARCHE ROBOTICS

**Arche Robotics** · 2020 – 2023 · Mechanical design engineer

AMR | Retail | Dashboards










### What?

- Promobot helps businesses **engage customers** and build their brand in physical and digital stores.
- Features: **welcoming**, **customer guidance**, **marketing** and **AR applications**.

### How?

- A **data-driven** companion that supports **digital transformation**.
- Monthly dashboards of sensor data such as **air quality** and **demographics**.

### Results

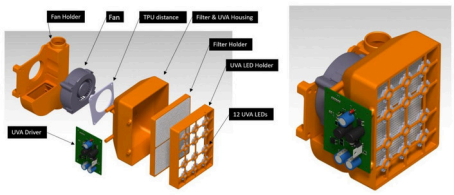
- **Deployed** at **MediaMarkt** Metropol Istanbul on 2 Jan 2023.
- Interacted with **400 customers** per month on average.
- Drove **10+ km** autonomously per month without mechanical issues.

## PURA

ARCHE ROBOTICS

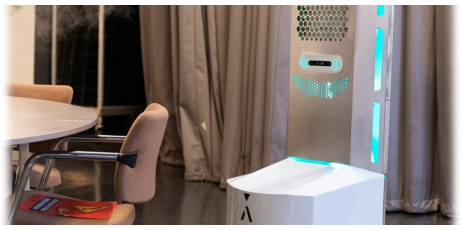
**Arche Robotics** · 2020 – 2023 · Mechanical design engineer

Photocatalysis | Air sensing







### What?

- **PURA** is based on the AT1 platform and **sanitizes** surfaces and floors and **purifies the air**.
- Designed a sanitation chamber with a **photocatalyst module** and **HOCl disinfection** liquid.
- Senses **air quality**: particles, VOC, humidity, temperature.

### How?

- Photocatalyst module with **16 ceramic foams** coated with $TiO_2$ and **40 UVA LEDs**.
- A **piezo fog** generator makes a **dry fog** of disinfectant, blown by a DC fan at **150 $m^3$/h**.
- No consumables such as HEPA filters, cartridges or mercury UVC bulbs.

### Results

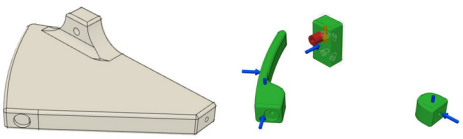
- **Deployed** an autonomous disinfection robot that works **24/7**.
- Cleans as much air per hour as **300 people** breathe.
- Works **without harmful chemicals**.
- Built strategic relationships with Chinese manufacturers.

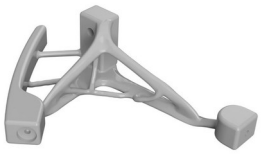
## GENERATIVE DESIGN STUDY

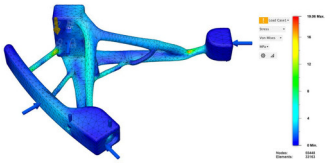
ARCHE ROBOTICS

**Arche Robotics** · 2020 – 2023 · Mechanical design engineer

Fusion 360 | Generative design







### What?

- Used **generative design** (AI-driven algorithms) to create **lightweight parts** that meet performance and design requirements.

### How?

- Used the **Fusion 360** generative design toolbox.
- Defined the **loads** on the part; compared **von Mises stress** results to ensure structural integrity.

### Results

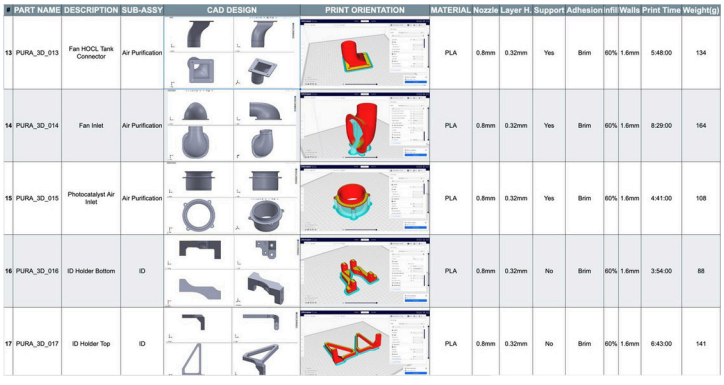
- **30% less weight** with all design constraints and performance criteria met.

## ADDITIVE MANUFACTURING PART MANAGEMENT

ARCHE ROBOTICS

**Arche Robotics** · 2020 – 2023 · Mechanical design engineer

FDM | Part tracking



| # | PART NAME | DESCRIPTION | SUB-ASSY | CAD DESIGN | PRINT ORIENTATION | MATERIAL | Nozzle | Layer H. | Support | Adhesion | infil | Walls | Print Time | Weight(g) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 13 | PURA_3D_013 | Fan HOCL Tank Connector | Air Purification | | | PLA | 0.8mm | 0.32mm | Yes | Brim | 60% | 1.6mm | 5:48:00 | 134 |
| 14 | PURA_3D_014 | Fan Inlet | Air Purification | | | PLA | 0.8mm | 0.32mm | Yes | Brim | 60% | 1.6mm | 8:29:00 | 164 |
| 15 | PURA_3D_015 | Photocatalyst Air Inlet | Air Purification | | | PLA | 0.8mm | 0.32mm | Yes | Brim | 60% | 1.6mm | 4:41:00 | 108 |
| 16 | PURA_3D_016 | ID Holder Bottom | ID | | | PLA | 0.8mm | 0.32mm | No | Brim | 60% | 1.6mm | 3:54:00 | 88 |
| 17 | PURA_3D_017 | ID Holder Top | ID | | | PLA | 0.8mm | 0.32mm | No | Brim | 60% | 1.6mm | 6:43:00 | 141 |

### What?

- Created an **Excel-based tracking system** to document 3D-printed parts and keep an organized **inventory**.

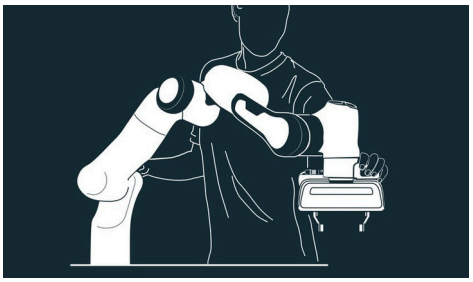
### Results

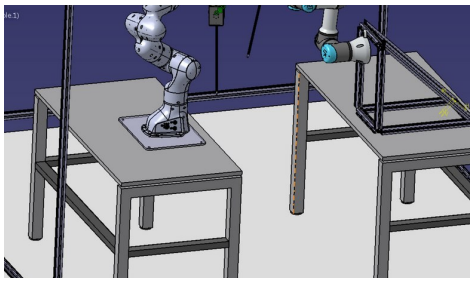
- **Standardized** part quality through identical slicing settings.
- **Efficient tracking** of series production, total print time and material use.

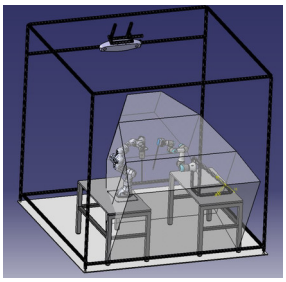
## SURGICAL ROBOT FOR AUTONOMOUS SUTURING

**OzU Robotics Lab** · Mechanical design

CATIA V5 | UR3 | Franka Panda







### What?

- A **surgical robot** with two cobots, **UR3** and **Franka Panda**. I designed the **end-effector flanges** for the Franka Panda arms and the structure for the **3D vision sensors**.

### How?

- Modeled robots and sensors in **CATIA V5**.
- Adapted the **Da Vinci EndoWrist** as end-effector.
- **Built** tables from welded rigid profiles to reduce vibration.

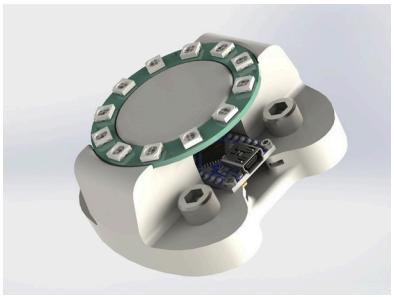
### Results

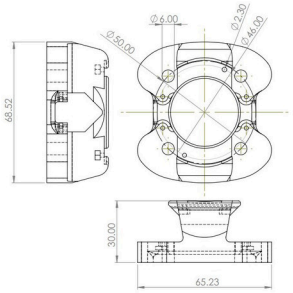
- Ongoing at the **OzU Robotics Laboratory**: "Development of Surgical Robot Assistants for Autonomous Suturing with Skill Transfer from Surgeon to Robot", funded by **TÜBİTAK 1001**.

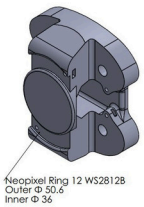
## KUKA KR20-3 LIGHT PAINTING

**Ozyegin University – RobArt** · 2021 – 2022

KUKA | Rhino/Grasshopper





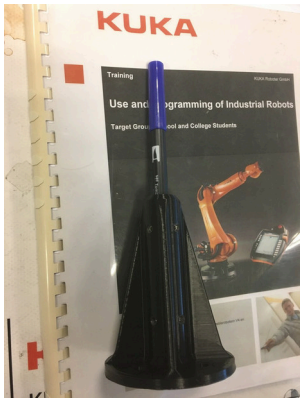

### What?

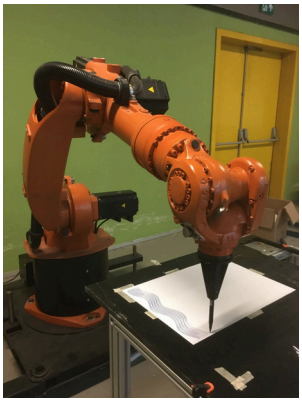
- In my senior year at Ozyegin University I took part in the **RobArt** workshops with industrial robot arms.
- **Light painting**: programmed a **KUKA KR20-3** with **Rhino/Grasshopper** and photographed the planned trajectory.
- **Designed and 3D printed** a flange holding an **LED ring**.

## KUKA KR20-3 PEN PLOTTER

**Ozyegin University – RobArt** · 2021 – 2022

KUKA | 3D printing



### What?

- A second RobArt workshop on **pen-plotter art**: the robot arm draws 2D images.
- **Designed and 3D printed** a flange holding a pen on the KUKA KR20-3.

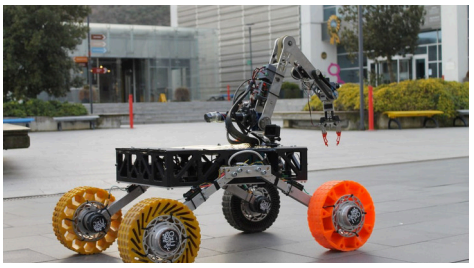
## THE HOPE

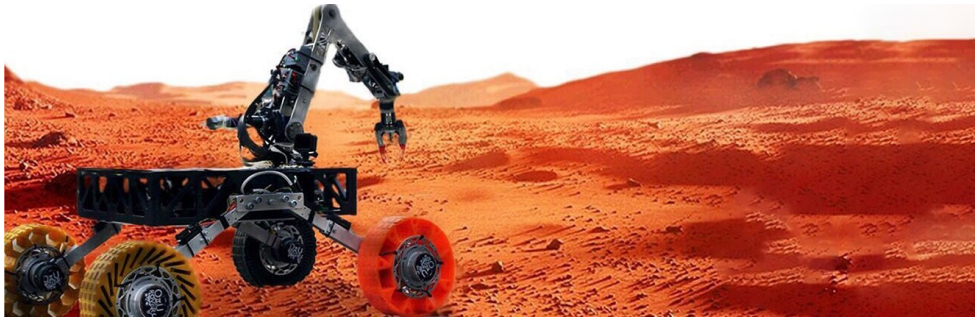
**OzU Rover Team** · 2019 – 2020 · Team lead

Mars rover URC









### What?

- **The Hope** is a Mars rover prototype built by the OzU Rover Team for the **University Rover Challenge 2020** in the USA.

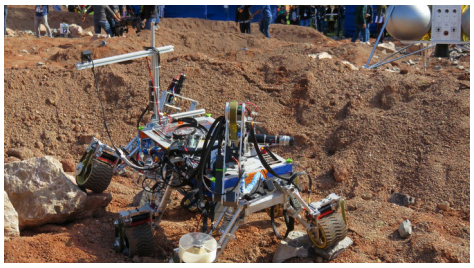
### Results

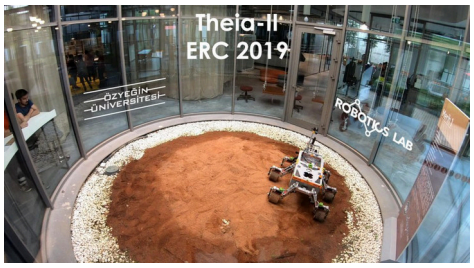
- **Qualified**, ranking in the top 36 of 102 teams.
- The event was cancelled due to COVID-19.

## THEIA

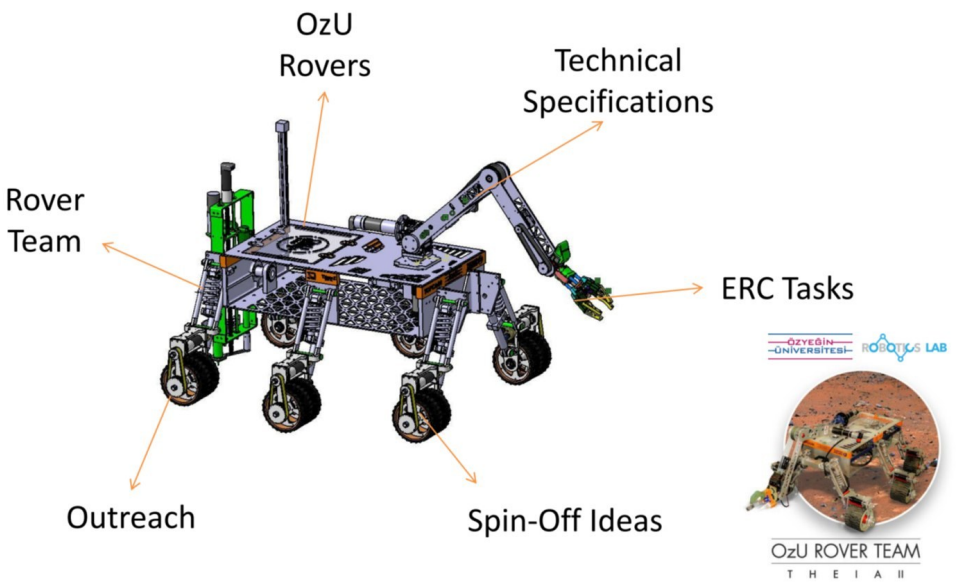
**OzU Rover Team** · 2018 – 2019 · Mechanical designer, then team lead

Mars rover ERC








### What?

- Theia was built by the OzU Rover Team for the **European Rover Challenge**.
- Teams design and build a **Mars rover prototype** for **tele-operated** or **autonomous** tasks that assist astronauts.

### Results

- **ERC 2018: 26th/65** as mechanical designer.
- **ERC 2019: 13th/56** as **team lead**, through leadership and design optimization.